Toulouse School of Economics, 2014-2015
M2 – Macroeconomics 1 – Professor Franck Portier

Exam

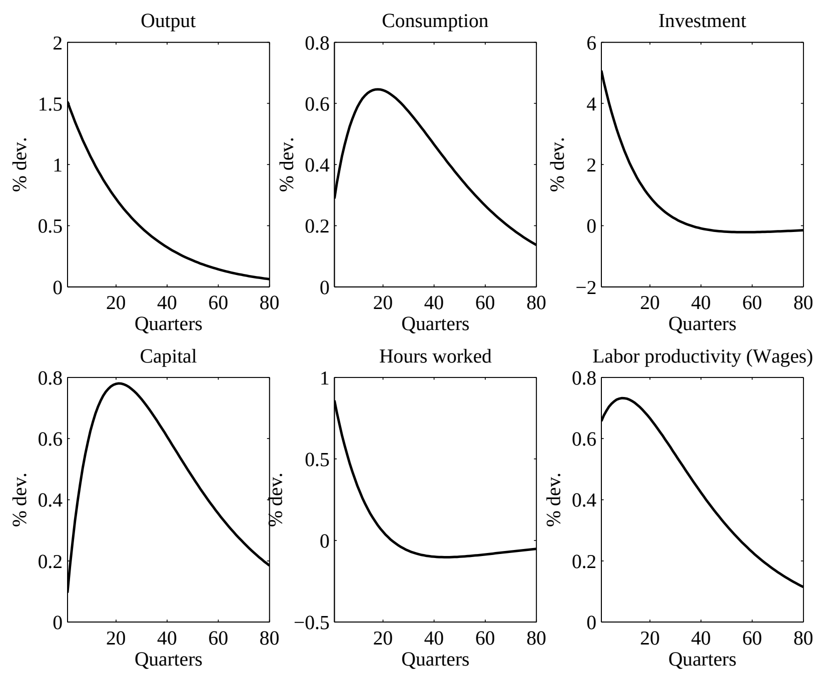
This is a 3 hours exam. Lecture slides and any handwritten material are allowed.
You must (please) write legibly.

I – Questions (*1/3 of the points*)

You should target around 15-20 lines per question. Copying the slides will not bring any points.

1 – Figure 1 below display the responses of some macroeconomic variables to a persistent but not permanent Total Factor Productivity shock in the simple RBC model of the course. Give the economic mechanisms behind those responses. You may want to use the equations that define a competitive equilibrium in that model.

Figure 1: Response to a persistent TFP shock



2 – In the Mortensen-Pissarides' model, the stock of vacancies is a state variable. True? False? Uncertain?

3 – Shocks to the capital stock cannot be the main source of fluctuations in a Real Business Cycle model. True? False? Uncertain?

4 – What is the role of the transversality condition is the Ramsey model?

II – The Harrod-Domar's model (*1/3 of the points*)

The model of Harrod (1939) and Domar (1949) can be seen as a version of the Solow model but with limited substitution between captal and labor. It produces a result of *instability* of balanced growth. This problem solves the Harrod-Domar's model.

Time is continuous. Aggregate production function is

$$Y(t) = \min\left(\frac{K(t)}{v}, \frac{L(t)}{\alpha}\right), \quad \nu > 0, \alpha > 0. \tag{1}$$

$Y(t)$, $K(t)$ and $L(t)$ are, respectively, aggregate output, the capital stock, and employment.

The saving rate is exogenous and assumed to be $0 < s < 1$; labor force grows exogenously at rate n_L. The other equations of the model are therefore:

$$\begin{aligned}
S(t) &= sY(t) && (2)\\
I(t) &= \delta K(t) + \dot{K}(t) && (3)\\
I(t) &= S(t) && (4)\\
\frac{\dot{L}(t)}{L(t)} &= n_L && (5)
\end{aligned}$$

1 – Draw the isoquants of the production function. Derive expressions for Y/K, Y/L, and K/L under the assumption that both production factors are fully employed. What happens if the actual K/L is less than v/α? What if it is larger than v/α?

2 – Show that, in order to maintain full employment of capital in the model, output and investment must grow at the so-called "warranted rate of growth" which is equal to $(s - \delta v)/v$.

3 – Show that, in order to maintain full employment of labour in the model, output and investment must grow at the so-called "natural rate of growth" which is equal to n_L.

4 – Derive the condition under which the economy grows with full employment of both factors of production. This is called the Harrod-Domar's condition. What happens if $(s - \delta v)/v$ is above or below n_L?

5 – Show that a Harrod-Domar's like condition also appears in the Solow's model. *Hint: The Solow model is described by equations (2) to (5) plus (1) with a general neoclassical production function $F(K, L)$. Write the model is per capita terms and derive a condition that involves K/L and n_L at the steady state.*

6 – Explain why the Solow's model does not suffer from the instability (or knife-edge stability) of the Harrod-Domar's model.

III – The Diamond's coconut model (*1/3 of the points*)

Let's consider the following environment. Time is continuous. There is a continuum of agents of mass one on an island. The island is composed of a forest and a beach. The forest is where coconut trees grow, while the beach is where people meet. Coconuts are the only good in the economy. Each agent has a linear utility $u(t)$. Consuming one coconut gives utility $y > 0$, while utility is 0 when the agent is not consuming. Agents discount utility at rate r (which is also the risk free rate as utility is linear) and the cost of collecting a coconut is $c(t)$. Agents' intertemporal expected utility, as evaluated at date T, is therefore

$$V(T) = E_T \int_T^\infty e^{-rt}\big(u(t) - c(t)\big)dt$$

Production is done by climbing at coconut trees, who are of various height. Agents without coconuts walk in the forest and look for trees. They are said to be *unemployed.* When they are unemployed, they randomly bump into coconut trees. The process of finding a tree is Poisson with rate a. The cost $c(t)$ of collecting one coconut is $i.i.d$, with cdf $G(c)$, defined over $[\underline{c}, \overline{c}]$, with $\underline{c} > 0$. One cannot collect or carry more than one coconut.

There is a taboo on this island, according to which one cannot eat self-collected coconuts. Therefore, once a coconut is collected, an agent has to walk on the beach carrying her coconut, and she will randomly meet another agent with a coconut. When walking in search for a trade partner, agents are said to be *employed*. When a meeting happens, the pair of agents that meets exchange their coconuts (one against one) and eat them. We denote by $e(t)$ (*employment*) the measure of agents that are holding a coconut and looking for a partner. The process of meeting someone else on the beach is Poisson with rate $b(e)$. It is assumed that $b'(e) > 0$.

The life of an agent can therefore be described as follows: once agents have a coconut, they simply walk on the beach until they meet another agent with a coconut, and trade. Without a coconut, they walk in the forest until they run into the coconut tree, and then they decide whether to collect the coconut given the cost.

We assume that agents strategy is a stationary policy $p(c, e)$ which determines a probability of collecting a coconut given its cost $c \in [\underline{c}, \overline{c}]$ as also a function of the current measure of traders looking for partners, $e \in [0, 1]$:

$$p : \quad [\underline{c}, \overline{c}] \times [0, 1] \to [0, 1]$$

A symmetric equilibrium is a policy p that is the best response to itself, in the sense that when all other agents use p, it is optimal for each of them to do so as well.

We will assume that in such an environment, the optimal policy is a reservation cost policy, i.e., $p(e(t), c) = 1$ for all $c \leq c^\star(t)$ and $p(e(t), c) == 0$ otherwise (where we write $c^\star(t)$ instead of $c^\star(e(t))$ to simplify notation).

We denote $V_E(t)$ the value of holding a coconut and looking for a trade partner (being employed) and $V_U(t)$ the value of looking for a tree (being unemployed).

1 – Why don't we have to care about the price of a coconut in a match? Why does it simplify the analysis?

2 – How to justify $b'(e) > 0$? Explain why this assumption is called "thick market externality"? Why 'externality"?

3 – Write the arbitrage equation that gives the flow value of being employed. *Hint: by arbitrage, the flow value of being employed should be equal to the expected gain in value of meeting a partner plus the time change of $V_E(t)$:*

$$\begin{aligned}\texttt{flow value } rV_E(t) = \quad & \texttt{probability of a meeting} \times \Big(\texttt{utility of consumption} + \\ & \texttt{change in value from employment to unemployment}\Big) \\ & + \dot{V}_E(t)\end{aligned} \tag{1}$$

4 – Explain why the flow value of unemployment is given by

$$rV_U(t) = a \int_{\underline{c}}^{c^\star(t)} (V_E(t) - V_U(t) - c)dG(c) + \dot{V}_U(t) \tag{2}$$

5 – Express the threshold $c^\star(t)$ as a function of $V_E(t)$ and $V_U(t)$.

6 – Given the reservation cost policy, write the law of motion of employment e, which is here the equivalent of the BEVERIDGE curve in MORTENSEN-PISSARIDES:

$$\dot{e}(t) = ... \tag{3}$$

7 – For the rest of the problem, we restrict to steady states of this economy, denoted $(e^\star, c^\star)$. From equations (1) and (2) and the answer to question **5**, derive a first implicit relation (A) between $c^\star$ and $e^\star$. Show that (A) defines a positively sloped locus in the plane $(e^\star, c^\star)$ when $y > c^\star$. Explain why this condition $y > c^\star$ must hold. Admit that this locus $c = c_A(e)$ is a concave function.

8 – From (3), derive a second implicit relation (B) between $c^\star$ and $e^\star$. Show that (B) defines a positively sloped locus in the plane $(e^\star, c^\star)$. Denote $c = c_B(e)$

9 – A steady state is an intersection of $c_A(e)$ and $c_B(e)$. Explain why $(0, \bar{c})$ is always a steady state.

10 – Draw (A) and (B) in a case of two interior steady states. Explain in words the reason of multiplicity.

11 – Can you Pareto rank an interior steady state and the zero steady state? Explain what is going on.

12 – Is the level of e suboptimally low in any interior steady state? *Hint: the algebra is quite tedious. Restricting to steady state welfare comparison, you would need to assume a situation in which everybody commits to use a policy $\widehat{p}$ such that $\widehat{p}(c, e^\star) = 1$ for all trees with $c \leq c^\star + \delta$ for δ positive and small. Denoting $V_U(\delta)$ and $V_E(\delta)$ the values in such an economy, you would need to show that $V_U'(\delta = 0)$ and $V_E'(\delta = 0)$ are both positive. Unless you have a lot of time left, just explain what is the answer likely to be.*